



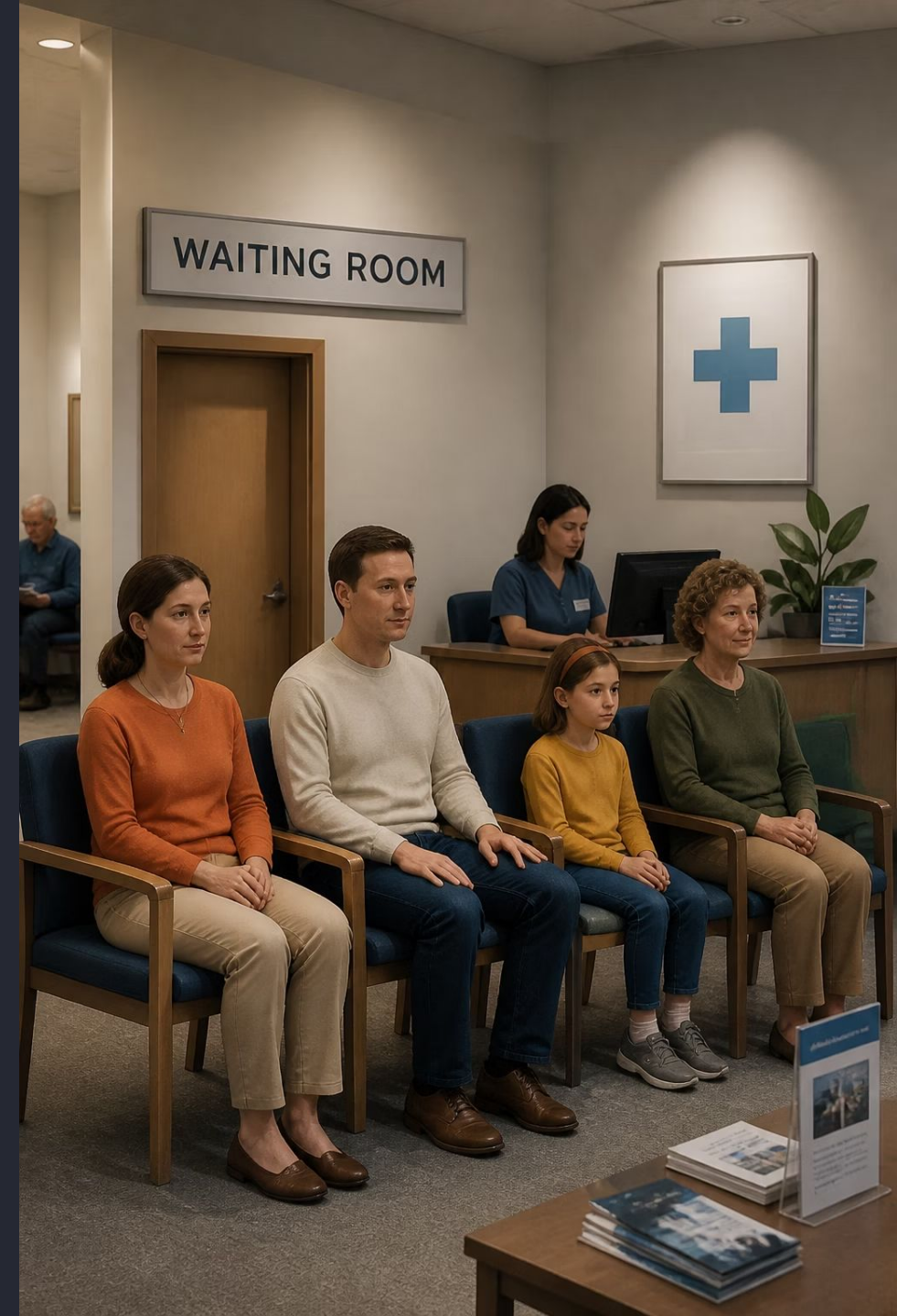
HealthConnect-AI

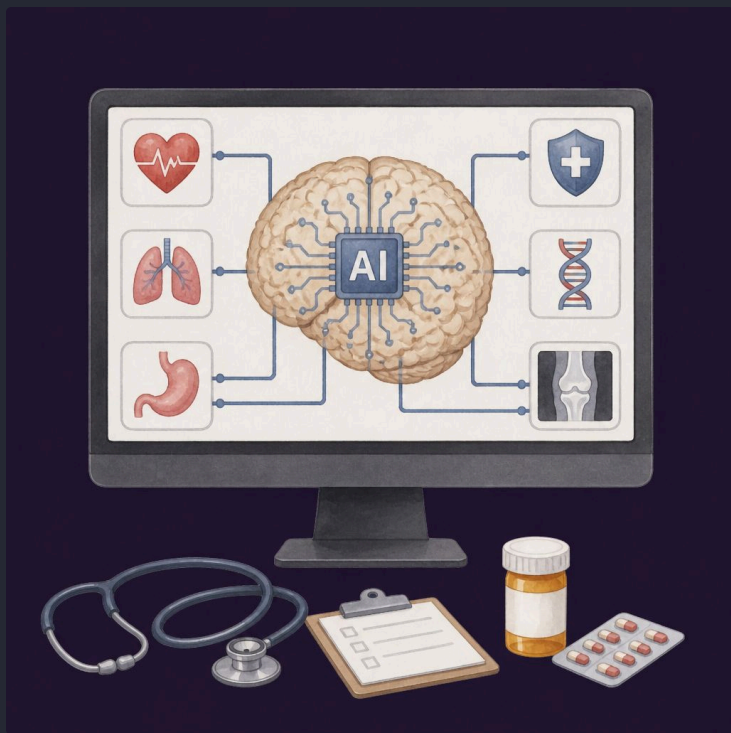
Intelligent Healthcare Platform Connecting Patients, Doctors & Hospitals

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Real-World Challenges in Healthcare

- 🕒 Delayed Pneumonia Diagnosis - Shortage of radiologists causes delayed diagnosis and increased risk of complications.
- 🏠 Limited Access to Healthcare - Rural areas lack specialists and diagnostic facilities, requiring long travel distances.
- 🧑 Manual X-ray Analysis - Radiologist interpretation is time-consuming and varies between experts.
- 📍 Difficulty Finding Nearby Hospitals - Patients struggle to locate hospitals with required specialists during emergencies.
- ⚡ Emergency Decision Making - Quick decisions are critical; delays worsen pneumonia outcomes.
- 👥 Healthcare Accessibility During Peak Hours - Overcrowding and limited availability information.





PNEUMO-AI Solution

One unified platform that transforms pneumonia detection into an intelligent, accessible diagnostic tool.

- **AI-Based Pneumonia Detection** using CNN to classify chest X-rays
- **Lung Segmentation** using U-Net model for improved accuracy
- **AI Health Report** with prediction results and confidence scores
- **Medical Specialist Recommendation** for pulmonologists
- **Nearby Hospital Search** using Google Maps integration
- **User-Friendly Interface** built with Streamlit

AI/ML Core Capabilities



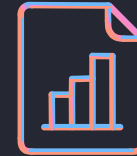
Deep Learning

Convolutional Neural Network (CNN) for accurate pneumonia classification from chest X-rays.



Computer Vision

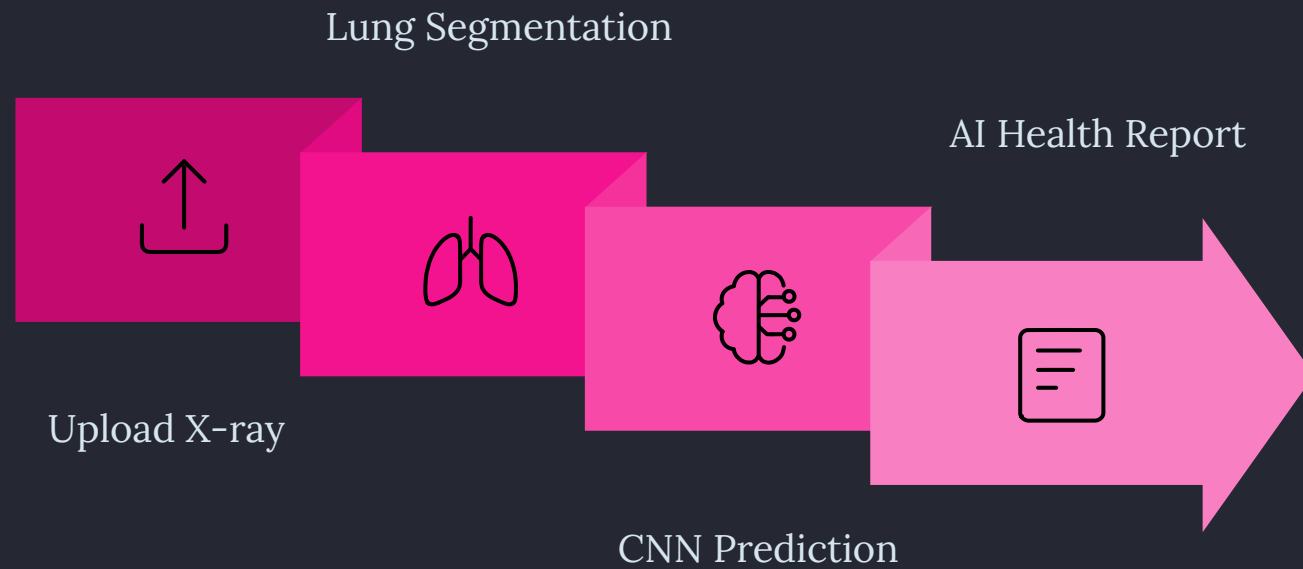
U-Net model for precise lung segmentation, improving prediction quality and interpretability.



Machine Learning

Automated health report generation with confidence scores and specialist recommendations based on AI analysis.

Patient Workflow



PNEUMO-AI guides patients from chest X-ray upload to hospital navigation with AI-powered analysis, recommendations, and next-step support.

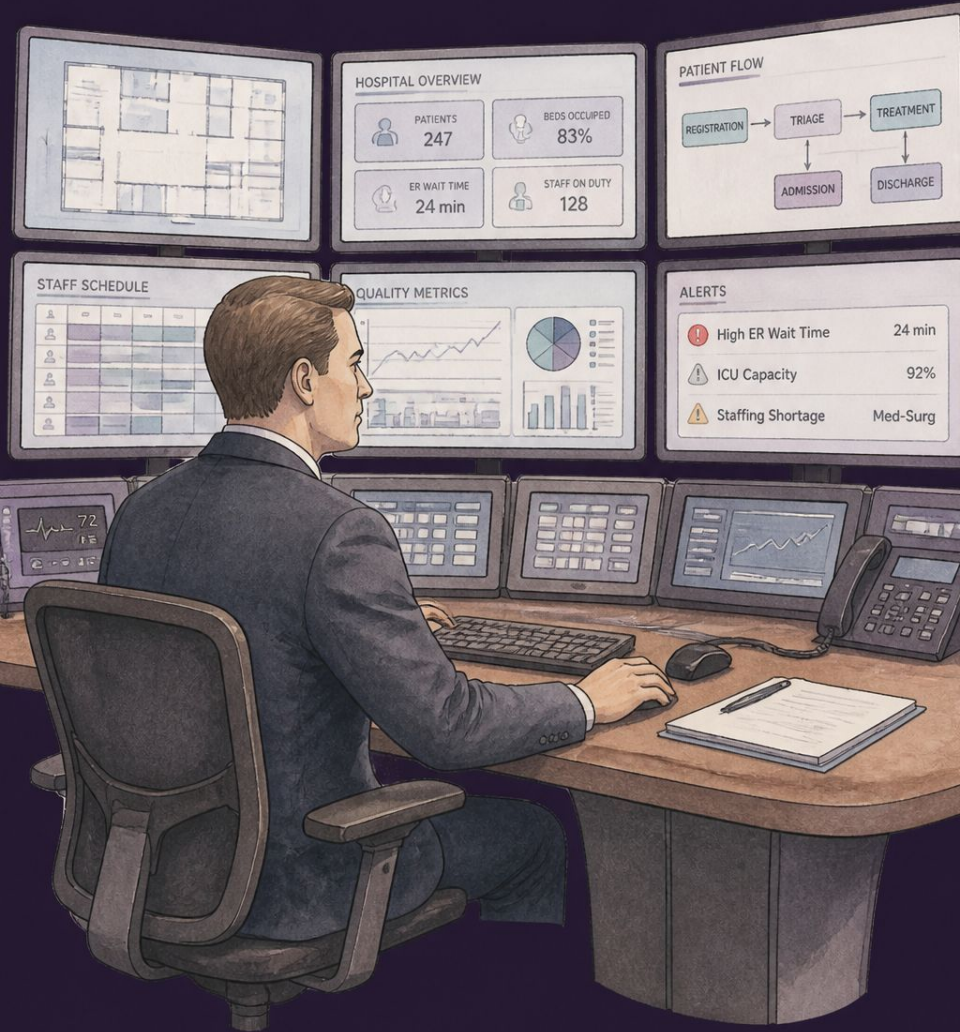
Doctor Dashboard

Designed around clinical workflows, the Doctor Dashboard puts AI-augmented insights at the center of every patient interaction.

- **Secure access** to authorized patient records and case histories
- **AI-analyzed results** — reviewed symptoms and X-rays with visual region highlights for rapid interpretation
- **Clinical decision tools** to document professional feedback and finalize diagnoses

□ AI augments clinical judgment — doctors remain in full control of every decision.





Platform Features

AI-Powered Detection

CNN-based pneumonia classification with high accuracy from chest X-ray images.

Intelligent Segmentation

U-Net model for precise lung region identification before analysis.

Hospital Integration

Google Maps Places API integration for finding nearby hospitals with directions, contact info, and website links.

Technology Stack



Backend

Python-based REST APIs with secure authentication.



AI/ML

CNN for pneumonia detection and U-Net for lung segmentation.



Database

Secure data management with encryption for patient records.



Frontend

Streamlit-based user-friendly interface for patients and healthcare professionals.



Integration

Google Maps Places API for hospital location services.

Key Benefits for Healthcare Professionals



Faster Diagnosis

AI-assisted pneumonia detection dramatically reduces time from X-ray upload to actionable diagnostic insight.



Explainable Results

Confidence scores and lung segmentation visualization build clinician trust in AI predictions.



Seamless Referrals

Automated specialist recommendations and integrated hospital search streamline patient care pathways.

Impact & Accessibility

Impact

- **Early pneumonia detection**
- **Faster AI-based diagnosis**
- **Improves patient care**

Accessibility

- **Web-based application**
- **Easy-to-use interface**
- **Cross-platform support**

Next Steps

01

Pilot Program

Launch with partner hospitals and clinics to validate AI performance in real clinical environments.

02

Regulatory Compliance

Pursue medical certifications and compliance frameworks relevant to each operating region.

03

Platform Scale

Expand connectivity to onboard more healthcare providers, specialists, and patients nationwide.



Thank You

